Continuing from last class

April 1, 2026

e.g. Compute $\iint_R x^2 + y \, dA$

for $R = [-1,2] \times [1,4] = \{(x,y) \in \mathbb{R}^2 : -1 \leq x \leq 2,\ 1 \leq y \leq 4\}$

$$\iint \cdots dx\,dy = \int_1^4 \left[\int_{-1}^2 (x^2+y)\,dx\right] dy = \int_1^4 \left[\frac{x^3}{3} + yx \Big|_{x=-1}^{x=2}\right] dy$$

integrating first with respect to x

$$= \int_1^4 \left(\frac{8}{3} + 2y\right) - \left(\frac{-1}{3} - y\right) \, dy = \int_1^4 3 + 3y \, dy$$

$$= \left[3y + \frac{3}{2} y^2 \Big|_{y=1}^{y=4}\right] = 12 + 24 - 3 - \frac{3}{2}$$

$$= 31.5$$

$$\iint \cdots dy\,dx = \int_{-1}^2 \left[\int_1^4 (x^2+y)\,dy\right] dx = \int_{-1}^2 \left[x^2 y + \frac{y^2}{2} \Big|_{y=1}^{y=4}\right] dx$$

integrating first with respect to y

$$= \int_{-1}^2 (4x^2 + 8) - \left(x^2 + \frac{1}{2}\right) dx = \int_{-1}^2 3x^2 + \frac{15}{2} \, dx$$

$$= x^3 + \frac{15}{2} x \Big|_{x=-1}^{x=2} = (8+15) - \left(-1 - \frac{15}{2}\right)$$

$$= 23 + 1 + 7.5 = 31.5$$

And how about splitting the sum $x^2 + y$:

$$\iint_R x^2 + y \, dA = \iint_R x^2 \, dA + \iint_R y \, dA$$

$$= \int_1^4 \left[\int_{-1}^2 x^2 \, dx\right] dy + \int_{-1}^2 \left[\int_1^4 y \, dy\right] dx$$

$$= \left(\int_{-1}^2 x^2 \, dx\right)\left(\int_1^4 1 \, dy\right) + \left(\int_1^4 y \, dy\right)\left(\int_{-1}^2 1 \, dx\right)$$

$$= \left(\frac{x^3}{3} \Big|_{x=-1}^{x=2}\right)\left(y \Big|_{y=1}^{y=4}\right) + \left(\frac{y^2}{2} \Big|_{y=1}^{y=4}\right)\left(x \Big|_{x=-1}^{x=2}\right)$$

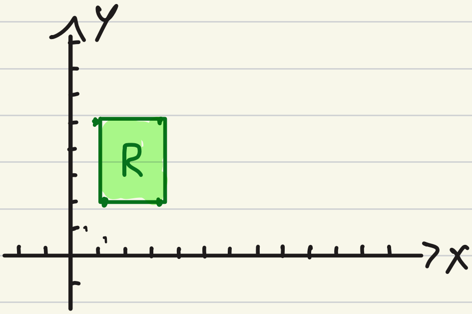
$$= \left(\frac{8}{3} - \frac{-1}{3}\right)(3) + \left(8 - \frac{1}{2}\right)(3)$$

$$= 9 + 24 - 3/2 = 31.5$$

Order of integration

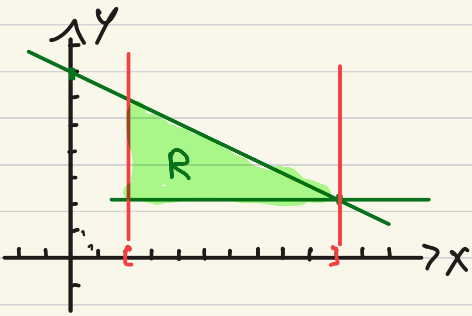
Let's sketch some regions: (Examples of elementary regions)

- $R = \{ (x,y) \in \mathbb{R}^2 : 1 \le x \le 3,\ 2 \le y \le 5 \} = [1,3] \times [2,5]$ rectangle

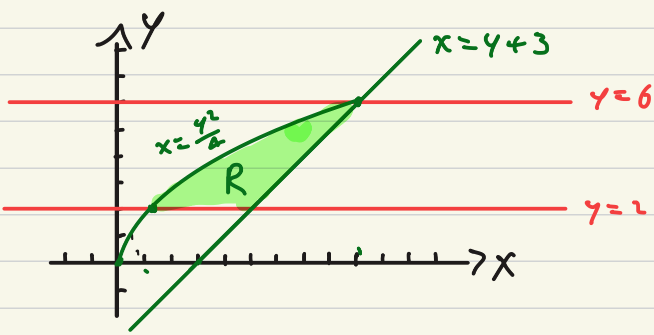


- $R = \{ (x,y) \in \mathbb{R}^2 : 2 \le x \le 10,\ 2 \le y \le 7 - \frac{x}{2} \}$

rectangular triangle with sides parallel to x and y axes



- $R = \{ (x,y) : 2 \le y \le 6,\ \frac{y^2}{4} \le x \le y+3 \}$



def: We say that a region $D \subseteq \mathbb{R}^2$ is:

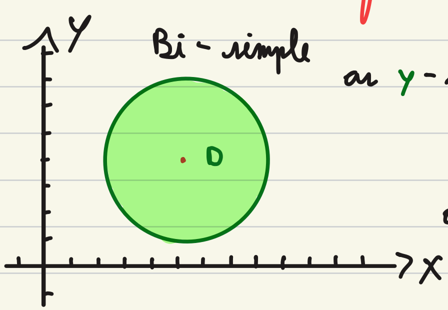
1) y-simple, if we can describe it as
$$D = \{(x,y) \in \mathbb{R} : a \leq x \leq b,\ \gamma(x) \leq y \leq \delta(x)\}$$
for some continuous functions $\gamma, \delta : [a,b] \to \mathbb{R}$

2) x-simple, if we can describe it as
$$D = \{(x,y) \in \mathbb{R} : a \leq y \leq b,\ \alpha(y) \leq x \leq \beta(y)\}$$
for some continuous functions $\alpha, \beta : [a,b] \to \mathbb{R}$

3) Bi-simple, if D is both x-simple and y-simple

4) elementary, if D is either x-simple or y-simple

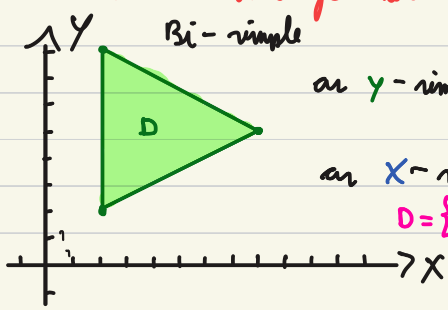
e.g. · D is the circle of radius 3 centered at $(5,4)$



as y-simple region $D = \{(x,y) \in \mathbb{R}^2 :$
$$2 \leq x \leq 8,$$
$$4 - \sqrt{9-(x-5)^2} \leq y \leq 4 + \sqrt{9-(x-5)^2}\}$$

as x-simple region
$$D = \{(x,y) \in \mathbb{R}^2 : 1 \leq y \leq 7,$$
$$5 - \sqrt{9-(y-4)^2} \leq x \leq 5 + \sqrt{9-(y-4)^2}$$

· D is the triangle with vertices $(2,2), (8,5), (2,8)$



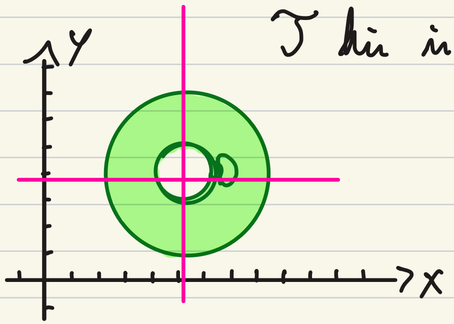
as y-simple region $D = \{(x,y) \in \mathbb{R}^2 : 2 \leq x \leq 8,$
$$\frac{x}{2} + 1 \leq y \leq -\frac{x}{2} + 9\ \}$$

as x-simple region $D =$ addendum after class

$$D = \{(x,y) \in \mathbb{R}^2 : 2 \leq y \leq 8,\ 2 \leq x \leq 8 - 2|5-y|\}$$

morale: not all simple descriptions are equally simple

· Example?



This is not an elementary region

Integrating over simple regions

1) **y**-simple, if we can describe it as

$$D=\{(x,y)\in \mathbb{R} : a\leq x\leq b,\ \gamma(x)\leq y\leq \delta(x)\}$$

for some continuous functions $\gamma, \delta : [a,b] \to \mathbb{R}$

$$\int_a^b \int_{\gamma(x)}^{\delta(x)} f \, dy \, dx$$

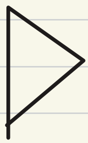
2) **x**-simple, if we can describe it as

$$D=\{(x,y)\in \mathbb{R} : a\leq y\leq b,\ \alpha(y)\leq x\leq \beta(y)\}$$

for some continuous functions $\alpha, \beta : [a,b] \to \mathbb{R}$

$$\int_a^b \int_{\alpha(y)}^{\beta(y)} f \, dx \, dy$$

e.g. $f(x,y) = x^2 y$ integrate over the triangle with vertices $(0,0), (2,2), (0,4)$



$$D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le 2, \; x \le y \le -x+4\}$$

$$\int_0^2 \int_x^{4-x} x^2 y \, dy \, dx$$

T.B.C.